Electricity Demand Forecasting

Contents

1. Abstract

In this study, we attempt to forecast the power demand for a short-term forecast horizon in Delhi, India, based on real-time weather conditions through accurate and reliable forecasting models employing MLR. The dataset was sourced from www.kaggle.com. It has 15 fields, including electric power demand (in kW) every 5 minutes and environmental measurements such as temperature, humidity, wind speed, wind direction, pressure, and the timestamp breakdown into year, month, day, hour and minute and 3-time-step moving average of power demand.

The model-development procedure in the study included data preparation, feature engineering, model building and finally, evaluation. Model-specified assumptions in MLR for linearity, stationarity, normality, homoscedasticity, and multicollinearity are evaluated and held to be sound for statistical robustness. The work concentrated on short-term electricity demand prediction in Delhi based on Multiple Linear regression (MLR) using current and lagged weather parameters (like Temperature, Humidity, and Wind speed) as independent variables and power demand as dependent variables. Three MLR models were created to compare the effect of temperature at different times: Model A had the temperature of the present; Model B lagged the 30-minute temperature; Model C lagged the 1-hour temperature. Although Model A showed the best statistical significance and lowest AIC, it was based on real-time temperature data, which may be unavailable for forecast purposes. On the contrary, a completely negative temperature coefficient was found in Model C.

Model B, with a non-significant temperature variable, was capable of almost the same prediction accuracy (R^2 = 99.67%, RMSE ≈ 79.26) and was based completely on lagged variables, making it more useful in practical implementation. After analysing, comparing and judging models in terms of model diagnostic tools, statistical validity and operational applicability, Model B was considered the best fit for short-term electricity demand prediction in Delhi. This study has underscored the importance of lagged weather variables and historical demand in creating reliable and operationally applicable models for the prediction of power demand.

2. Introduction

The growing uncertainty on urban electricity demand due to changing weather patterns underscores the importance of precise and timely forecasting of electricity demand (Singh et al., 2020). Efficient power demand forecasting models are crucial to the stability of energy systems, especially in rapidly growing urban areas such as Delhi. Accurate forecasting enables utility companies to better optimize generation schedules, minimize energy waste, and avoid shut-offs, ultimately making the energy industry more sustainable and efficient (Chapagain & Kittipiyakul, 2018).

In the past, electricity consumptions were forecasted according to time series and time patterns. Nevertheless, in the last years, weather variables (temperature, humidity, wind speed, etc.) have been increasingly identified as influencing short-term energy use. High temperatures could lead to the use of air conditioning and cause load demand to rise, while wind speed and humidity could change thermal comfort and introduce changes in residential and commercial power consumption.

Urban centres such as Delhi are especially vulnerable to such climate change-induced shifts since they house a large proportion of the population, closely packed buildings and higher use of cooling technology during adverse weather events (Kumari et al., 2021). Therefore, the assessment and quantification of the impact made by weather variables on electricity demand is important for designing energy-resilient smart grids and energy-efficient urban infrastructures.

3. Literature Review

Recent reports have emphasized the role of meteorological factors on urban electricity consumption, as seen in Delhi, among other cities. For example, a study from Harish et al. (2020) observed an increase of 30% in the overall electricity demand in Delhi above 30°C, which implies that there is a significant relationship between the temperature and electricity demand (Harish et al., 2020). Meanwhile, the Institute for Energy Economics and Financial Analysis (IEEFA) noted that on hot and humid days, Delhi's peak power demand increased 3.8 times faster than on moderate days, emphasising the combined impact of temperature and humidity on electricity consumption (IEEFA, 2024).

Other meteorological factors, beyond temperature and humidity and other weather variables like wind speed, have been investigated for their impact on electrical consumption. In an article in the International Journal of Scientific and Research Publications, the demand for electricity in Delhi was forecasted using multiple regression analysis with temperature, humidity and precipitation as parameters, revealing the complex effect of various weather-related parameters (Goel & Goel, 2014).

In addition, the influence of the Urban Heat Island resulting from urbanization can lead to an increase in electricity use. Research by Kumari et al. (2021) notes that the urban region is warmer in comparison with the rural region, and this points to the fact that the cooling demand is higher, and electricity demand is on the high side (Kumari et al., 2021).

These results emphasize the need for using several weather parameters in forecasting models of electricity demand. While model forms such as MLR can provide an understanding of linear relationships, higher-power machine learning methods like Random Forest and Gradient Boosting may account for nonlinear interactions among predictors, yielding better and more accurate forecasts.
Finally, the literature highlights that the weather variables play a key role in determining urban energy demand. For a megacity such as Delhi, whose variability and extremes of the weather are on the rise, it is important to have robust models for predicting to make the best use of various meteorological factors for planning and management of energy.

3.1 Problem Identified

However, while weather-sensitive forecasting models have gained importance, no unanimous consensus has been reached in the literature on what meteorological variables contribute the most to short-term electricity consumption (Rick & Berton, 2022). Most previous studies ignore lagged weather variables, or they do not localize these models to represent specific climate patterns and urban behaviours (Jain & Gupta, 2024). Additionally, cities such as Delhi, characterized by rapid urbanization, high population density, and disturbing climate scenarios, require finer-grained, location-specific demand prediction models.

Most traditional models are temperature-centric, and they might possibly disregard the combined effect of other dependent variables, such as humidity and wind speed, that have resulted in playing a significant role in shaping the behaviours of the cooling and heating load (AL-Musaylh et al., 2019). Furthermore, the immediate and interactive impact of lagged meteorological characteristics, such as how weather from previous hours keeps affecting current electricity consumption, are commonly underestimated.

This research attempts to cover these gaps by establishing a data-driven, localized short-term demand prediction model for Delhi based on Multiple Linear Regression (MLR). It aims to investigate how the short-term weather variables, namely the temperature, humidity, wind speed and the lagged effect of all the mentioned factors, will contribute to the prediction of power demand. The study aims to provide practical suggestions for urban planners, energy utilities, and policymakers by assessing the model prediction accuracy and variable importance. This knowledge can be used to enhance energy management practices, energy efficiency efforts and smart grid advancement in meteorological dependent urban environments.

3.2 Lagged Weather Variables

These lagged temperature, humidity and wind speed variables are necessary in MLR models for the forecast of short-term electricity demand, especially in cities such as Delhi, where weather can change dramatically within a few hours. Lagged variables are employed to measure the lagged effect of weather on electricity demand. For instance, the temperature at some time in past (i.e. 30 minutes ago) could continue to affect electricity consumption at the present time due to behavioural reactions (e.g., putting the air-conditioner on). Short-term electricity demand prediction models which make use of recent weather observations, outperform the models that do not incorporate these recent observations (Chapagain & Kittipiyakul, 2018).

The motive is that consumers' feedback to weather/temperature, particularly for cooling and heating, does not occur due to dislocation but with a certain delay due to, for instance, building heat inertia and appliance usage. Models including weather at more than 1 lag achieved higher prediction accuracies, especially when the temperature changes abruptly within the day (Staffell & Pfenninger, 2018).

Humidity and the speed of the wind also play a major role. Moisture and wind likely affect thermal comfort and, therefore, the use of HVAC (Heating, Ventilation, and Air Conditioning). Lagged features of these variables enable models to capture their accumulated impacts on electricity demand over time.
In our work, we employ lagged features (e.g. temperature_lag30min, temperature_lag1hr, humidity_lag30min, humidity_lag1hr, windspeed_lag30min, windspeed_lag1hr) to enhance the performance of short-term demand forecasting by MLR for a typical urban environment of Delhi.

4. Data Preparation

4.1 Data Cleaning and Feature Engineering

The dataset was pre-processed before the start of modelling. Column names were modified to facilitate understanding the data and maintain consistency (power demand to power_demand, temp to temperature, wspd to wind_speed). The datetime column was cast to Date for easier handling of time-related features. Missing values were then determined by column-wise NA checks. There were two rows with 2 missing values for the moving_avg_3 column and 540 for the wind_dir column. The rows with any missing values (na) were dropped using na.omit(). to maintain the integrity of the models.

```
colSums(is.na(df))
  datetime power_demand  temperature     dew_point      humidity      wind_dir
         0            0            0             0             0           540
wind_speed     pressure         year         month           day          hour
         0            0            0             0             0             0
    minute   moving_avg
         0            2
```

Fig 1: Missing Values Columns

After the cleaning process, we generated new lagged weather variables to represent the lagging influence of temperature, humidity and wind speed on electricity demand. In detail, lagged values of temperature, humidity and wind_speed with 30 minutes and hourly were calculated using the lag() function. This then added further missing values back at the top of the data set (due to lagging), which were also removed using a filter() step. The resulting dataset was now missing value-free and augmented with the engineered lag variables, so we could move to the next stage of modelling.

```
# 5-minute data: 6 lags = 30 min, 12 lags = 1 hour
# Create lagged variables using dplyr::lag()
df <- df %>%
  mutate(
    temperature_lag30min = lag(temperature, 6),
    temperature_lag1hr = lag(temperature, 12),
    humidity_lag30min = lag(humidity, 6),
    humidity_lag1hr = lag(humidity, 12),
    windspeed_lag30min = lag(wind_speed, 6),
    windspeed_lag1hr = lag(wind_speed, 12)
  )
# Check missing values
colSums(is.na(df))
          datetime          power_demand           temperature             dew_point
                 0                     0                     0                     0
          humidity              wind_dir            wind_speed              pressure
                 0                     0                     0                     0
              year                 month                   day                  hour
                 0                     0                     0                     0
            minute            moving_avg  temperature_lag30min   temperature_lag1hr
                 0                     0                     6                    12
 humidity_lag30min       humidity_lag1hr    windspeed_lag30min       windspeed_lag1hr
                 6                    12                     6                    12
```

Fig 2: Lagged Weather Variables

4.2 Exploratory Data Analysis

Descriptive summary statistics for each data set variable are presented in Figure 3. Skewness was analysed to check the normality distribution of all variables in the model. Skewness was computed rather than using box plots for the purpose of this evaluation because it quantitatively estimates the degree of asymmetry and for better objective-based decisions of transformation requirements rather than subjective visual interpretation.

```
> summary(df)
    datetime                    power_demand    temperature        dew_point          humidity
 Min.   :0001-01-20     Min.   :1302   Min.   : 4.00   Min.   :-8.60   Min.   :  5.0
 1st Qu.:0008-07-20     1st Qu.:3078   1st Qu.:20.00   1st Qu.:10.40   1st Qu.: 44.0
 Median :0016-03-20     Median :3834   Median :27.00   Median :15.40   Median : 67.0
 Mean   :0016-04-02     Mean   :3963   Mean   :25.54   Mean   :16.34   Mean   : 63.4
 3rd Qu.:0023-10-20     3rd Qu.:4872   3rd Qu.:31.00   3rd Qu.:23.90   3rd Qu.: 84.0
 Max.   :0031-12-20     Max.   :8632   Max.   :46.40   Max.   :30.30   Max.   :100.0
    wind_dir          wind_speed         pressure             year            month
 Min.   :  0.0    Min.   : 0.000   Min.   : 989.6   Min.   :2021   Min.   : 1.00
 1st Qu.: 50.0    1st Qu.: 5.400   1st Qu.:1003.0   1st Qu.:2021   1st Qu.: 3.00
 Median :160.0    Median : 7.600   Median :1009.0   Median :2023   Median : 6.00
 Mean   :163.8    Mean   : 7.865   Mean   :1008.9   Mean   :2022   Mean   : 6.45
 3rd Qu.:270.0    3rd Qu.:11.200   3rd Qu.:1015.0   3rd Qu.:2023   3rd Qu.: 9.00
 Max.   :360.0    Max.   :63.000   Max.   :1027.0   Max.   :2024   Max.   :12.00
      day               hour             minute           moving_avg     temperature_lag30min
 Min.   : 1.00    Min.   : 0.00    Min.   : 0.00    Min.   :1308    Min.   : 4.00
 1st Qu.: 8.00    1st Qu.: 5.00    1st Qu.:10.00    1st Qu.:3079    1st Qu.:20.00
 Median :16.00    Median :11.00    Median :25.00    Median :3834    Median :27.00
 Mean   :15.75    Mean   :11.19    Mean   :27.35    Mean   :3963    Mean   :25.54
 3rd Qu.:23.00    3rd Qu.:17.00    3rd Qu.:40.00    3rd Qu.:4871    3rd Qu.:31.00
 Max.   :31.00    Max.   :23.00    Max.   :55.00    Max.   :8598    Max.   :46.40
 temperature_lag1hr humidity_lag30min humidity_lag1hr windspeed_lag30min
 Min.   : 4.00      Min.   :  5.0     Min.   :  5.0   Min.   : 0.000
 1st Qu.:20.00      1st Qu.: 44.0     1st Qu.: 44.0   1st Qu.: 5.400
 Median :27.00      Median : 67.0     Median : 67.0   Median : 7.600
 Mean   :25.54      Mean   : 63.4     Mean   : 63.4   Mean   : 7.865
 3rd Qu.:31.00      3rd Qu.: 84.0     3rd Qu.: 84.0   3rd Qu.:11.200
 Max.   :46.40      Max.   :100.0     Max.   :100.0   Max.   :63.000
 windspeed_lag1hr
 Min.   : 0.000
 1st Qu.: 5.400
 Median : 7.600
 Mean   : 7.865
 3rd Qu.:11.200
 Max.   :63.000
```

Fig 3: Descriptive Summary Statistics

Variable	Skewness
power_demand	0.26
temperature	-0.33
dew_point	-0.05
humidity	-0.36
wind_dir	-0.14
wind_speed	0.64
pressure	-0.01
year, month, day, hour, minute	0

moving_avg	0.26
temperature_lag30min	-0.33
humidity_lag30min	-0.36
windspeed_lag30min	0.64

Table 1: Skewness of the Variable

Table 1 shows the skewness of most of the variables (power_demand, temperature, dew_point, humidity, wind_dir, pressure and their respective lagged values) were close to zero, which means approximately symmetric distributions for them. Both wind_speed and windspeed_lag30min showed a moderate positive skew (0.64), which often indicates that transformation, such as logarithmic or square root, should be considered. Nonetheless, since the skewness is still within reasonable limits, no variable has been transformed.

4.3 MLR Assumptions Check

Linearity and Homoscedasticity Assumption

Both the linearity and homoscedasticity can be checked with a residual plot as shown below in Fig. 4. If the residuals bounce randomly around the horizontal y = 0 line, then the assumption of linearity is probably met. Furthermore, a uniform distribution of residuals along all levels of predictions is a check of homoscedasticity, meaning the variance in errors is constant.

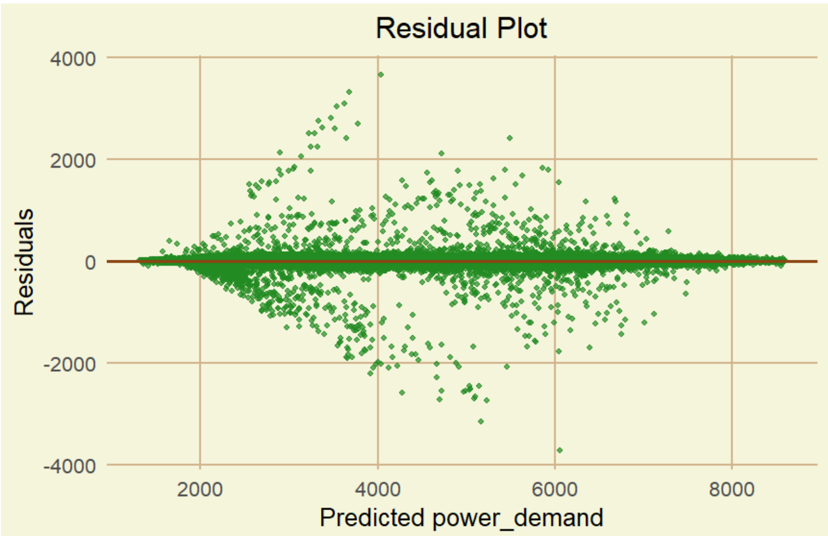

Fig 4: Residual Plot

The model residuals seem to cluster around zero, suggesting linearity. Although the residual plot showed some small minor patterns and some divergence in spread, no evident fan pattern existed, meaning weak heteroscedasticity. At a large enough sample size, the slight non-adherence of the data to its model assumptions (e.g., homoscedasticity) has little effect on the estimation or prediction of a model, an observation referred to as asymptotic robustness. So, we can declare minor heteroscedasticity as too mild to play any role in the real world when we use the model for prediction and not inference when the performance is also validated on a test set (e.g. RMSE or R^2) if other issues in the model like multicollinearity have been addressed.

Autocorrelation Assumption / Independence of Residuals

The Durbin-Watson statistic is a commonly used measure to detect the presence of autocorrelation in the residuals of a regression model. Table 2 below presents the results of the Durbin-Watson test performed using R.

Lag	D-W Statistics	P-value
1	1.0904	<2.2e-16

Table 2: Durbin-Watson Statistic
Alternative hypothesis: true autocorrelation is greater than 0

A Durbin-Watson (DW) test of 1.0904 and its corresponding p-value (< 2.2e-16) suggest positive autocorrelation in the residuals of the regression model and thus present evidence of positive autocorrelation. This is, being the main assumption on behalf of the independence of residuals, indicating that the model errors are not random but found instead to show a systematic behaviour throughout time. Time series data like electricity demand, in which the current state is conditioned on past trends or cycles, have this kind of autocorrelation. This problem can result in inefficient estimates and incorrect inferences, necessitating modifications to the model, such as including lagged variables.

Fix Autocorrelation

One way to deal with the problem of autocorrelation in regression-based models is to include lagged dependent variables, also referred to as autoregressive terms. This is done by including previous values of the dependent variable (e.g. power demand) as predictors of the model.

```
> # Include Lagged Dependent Variables (Autoregressive Terms)
> df <- df %>%
+   mutate(
+     powerdemand_lag1 = lag(power_demand, 1),
+     powerdemand_lag2 = lag(power_demand, 2)
+   ) %>%
+ filter(!is.na(powerdemand_lag1), !is.na(powerdemand_lag2))
```

Fig 5: Autoregressive Terms

After doing this, the Durbin-Watson test was re-applied to see if we had successfully offset the autocorrelation in the residuals.

Lag	D-W Statistics	P-value
1	2.0053	0.9507

Table 3: Updated Durbin-Watson Statistic
Alternative hypothesis: true autocorrelation is greater than 0

The revised Durbin-Watson test value for the revised regression model gives a DW statistic of 2.0053 and a p-value of 0.9506. This means that the residuals are no longer autocorrelated as DW is around 2, and the high p-value means there is insufficient evidence to reject the null of no autocorrelation. This gain implies that the incorporation of the lagged dependent variables demand_lag1 and demand_lag2 through regression successfully coped with the autocorrelation problem. Thus, the model is more appropriate for inference and forecasting since the most important assumption of residual independence is finally met.

Multicollinearity Assumption

Variable	VIF
temperature	**106.9**
humidity	**28.7**
wind_speed	4.8
temperature_lag30min	**210.4**
temperature_lag1hr	**106.3**
humidity_lag30min	**55.5**
humidity_lag1hr	**28.7**
windspeed_lag30min	8.9
windspeed_lag1hr	4.8
powerdemand_lag1	**318.6**
powerdemand_lag2	**318.6**

Table 4: VIF Statistic

The Variance Inflation Factor analysis shows there is serious multicollinearity between some predictors and current and lagged variables. For example, temperature (VIF = 106.9), temperature_lag30min (VIF = 210.4), as well as powerdemand_lag1 and lag2 (both VIF = 318.6) are highly correlated with one another (very high VIF), meaning that the information provided by them is redundant. There are also similar patterns for humidity and wind speed. Such multicollinearity can lead to interpretation issues with model parameters and increase the variance of the coefficient. To overcome this issue, but preserve interpretability, we select only the most influential lag variables in terms of their pairwise correlations and drop redundant lags but identities important lags.

Feature Selection

The selection of weather variables plays an important role in the development of a strong MLR model to forecast short-term electricity demand in Delhi. Although our correlation analysis revealed in Figure 6 shows that the current values of temperature, humidity, and wind speed are slightly stronger, or at least equally strong with power demand, we deliberately chose the 30-minute delayed values temperature_lag30min, humidity_lag30min, and windspeed_lag30min because they are more closely related, theoretically and practically, with the dynamics of electricity demand.

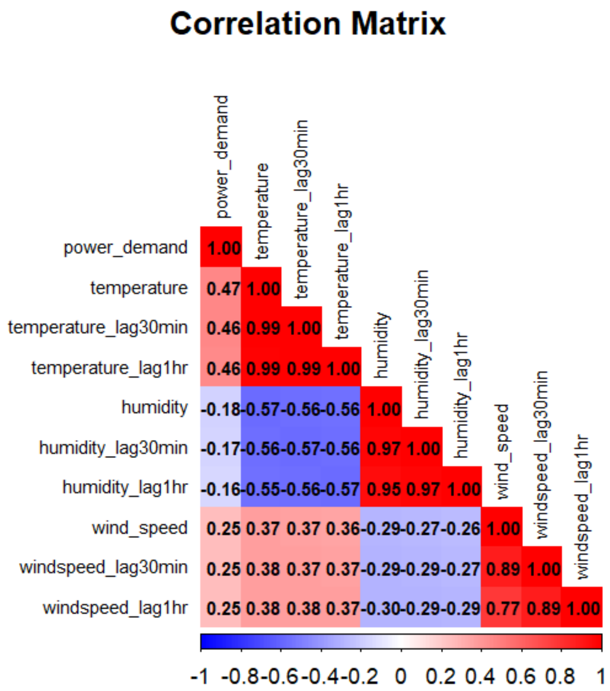

Fig 6: Correlation Matrix

Electric demand does not react immediately to weather conditions. Instead, there is typically a delay between the environmental stimulus and human behaviour. When people experience some type of discomfort, such as it being too hot, they often do not automatically adjust temperature systems, such as turning on the air conditioning immediately. Likewise, if the wind is blowing at a high speed, the indoors may feel colder than it is, and thus delay turning on the heating. These natural behavioural lags provide an argument for using lagged weather variables in forecasts, as this more accurately models the temporal link between weather changes and consequent power use.

Furthermore, the use of lagged variables also enforces causality, which might be more consistent with operational real-time forecasting where future or current weather is not known. By incorporating lagged weather features, we ensure the exogenous of the predictors, that is, they occur prior to the dependent variable power demand.

Both lagged and current weather variables can induce multicollinearity and result in unstable coefficient estimates. Our correlation test indicated that both the current and lag variables were highly correlated (above 0.95), and the VIF test also showed evidence of multicollinearity when both were in the model.

The selection of 30-minute lagged weather variables is motivated by the need to capture delayed demand effects, the practical limitations on forecasting, minimum multicollinearity between variables, and maintaining model stability. This is consistent with common practices in the industry and improves the prediction performance of electricity demand forecasts.

Splitting Datasets into training and testing datasets

In this study, the dataset was split into training and testing sets using stratified sampling to ensure a balanced distribution of power demand patterns across both subsets. An 80-20 split was applied, resulting in 314,307 observations for training and 78,577 for testing. Stratified sampling helps maintain the proportion of key variables, such as time-of-day or seasonal patterns, which is crucial for building a robust forecasting model that generalizes well to unseen data.

5. Model Building

The dataset used in this study is comprised of one numerical dependent variable, i.e., power demand, and multiple weather-related independent variables. According to the correlation analysis, the 30-min lagged values of temperature, humidity, wind speed and one lag of the power demand (powerdemand_lag1) were used as predictors, for their excellent correlation with the target variable and forecasting applicability. Temperature_lag30min, humidity_lag30min and windspeed_lag30min were selected to express the delayed response of environmental factors to electricity consumption, which is in line with previous studies and consumers' behaviours. Powerdemand_lag1 accounts for the dependence overtime on the consumption of electricity, which is often a feature of time-series data. There was no such collateral duplication between the selected predictors, and the variance inflation factor (VIF) values were all in a reasonable range. The resulting multiple linear regression (MLR) model seeks to explain the cause-and-effect relationship between electricity demand and influencing weather and historical demand variables.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-1.0975	0.8281	-1.3250	0.1850
temperature_lag30min	0.0161	0.0217	0.7410	0.4590
humidity_lag30min	0.0859	0.0064	13.2220	0.0000
windspeed_lag30min	0.2187	0.0240	9.1020	0.0000
powerdemand_lag1	0.9982	0.0001	8608.7560	0.0000

Table 5: MLR Regression Output

In constructing the MLR model to develop a short-term electricity demand forecast, temperature_lag30min was added first as the predictor together with other significant factors (humidity_lag30min, windspeed_lag30min and powerdemand_lag1). Although the model performed well (adjusted $R^2 = 0.9966$; Residual SE: 71.29), temperature_lag30min was found to be non-significant ($p = 0.459$). This led to further study on the appropriate shape of the temperature variable. We systematically compared three models, all based on three types of temperature (current, 30-minute lagged, and 1-hour lagged) with the other predictors fixed. We aimed to measure which temperature variable is most suitable to represent later lagged effects of weather on electricity demand, and in this way, made the final one both statistically robust and relevant to prediction.

Model Comparison

We considered here three alternative formulations of the MLR model that include different parametrizations of the temperature and holding the predictor terms such as humidity_lag30min, windspeed_lag30min and powerdemand_lag1 constant. For model A, the current temperature was used; for model B, the temperature with a 30-minute lag; and for model C, the temperature with an hour lag. Both statistically and from a practical view, the temperature variable in Model A was very significant ($p=3.47e-07$), whereas in Model B, there was no statistical significance ($p=0.459$). Model C had a marginally significant temperature ($p = 0.0199$) with a negative coefficient, which contradicted its expectations, particularly in warm climate cities like Delhi, where electricity demand usually increases with temperature.

```
> summary(model_A)

Call:
lm(formula = power_demand ~ temperature + humidity_lag30min +
    windspeed_lag30min + powerdemand_lag1, data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-4719.1   -20.4    -3.2    19.8  4970.1

Coefficients:
                     Estimate Std. Error  t value Pr(>|t|)
(Intercept)        -3.4967373  0.8197824   -4.265 2.00e-05 ***
temperature         0.1103702  0.0216764    5.092 3.55e-07 ***
humidity_lag30min   0.1010605  0.0064460   15.678  < 2e-16 ***
windspeed_lag30min  0.1982271  0.0240503    8.242  < 2e-16 ***
powerdemand_lag1    0.9980873  0.0001163 8581.746  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 71.28 on 314292 degrees of freedom
Multiple R-squared:  0.9966,    Adjusted R-squared:  0.9966
F-statistic: 2.324e+07 on 4 and 314292 DF,  p-value: < 2.2e-16
```

Fig 7: Model A MLR Output in R

```
> summary(model_B)

Call:
lm(formula = power_demand ~ temperature_lag30min + humidity_lag30min +
    windspeed_lag30min + powerdemand_lag1, data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-4718.9   -20.4    -3.2    19.8  4971.0

Coefficients:
                      Estimate Std. Error  t value Pr(>|t|)
(Intercept)          -1.093187   0.828155   -1.320    0.187
temperature_lag30min  0.016045   0.021769    0.737    0.461
humidity_lag30min     0.085915   0.006498   13.222   <2e-16 ***
windspeed_lag30min    0.218732   0.024039    9.099   <2e-16 ***
powerdemand_lag1      0.998284   0.000116 8608.544   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 71.29 on 314292 degrees of freedom
Multiple R-squared:  0.9966,    Adjusted R-squared:  0.9966
F-statistic: 2.324e+07 on 4 and 314292 DF,  p-value: < 2.2e-16
```

Fig 8: Model B MLR Output in R

```
> summary(model_C)

Call:
lm(formula = power_demand ~ temperature_lag1hr + humidity_lag30min +
    windspeed_lag30min + powerdemand_lag1, data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-4718.7   -20.4    -3.1    19.8  4971.8

Coefficients:
                     Estimate Std. Error  t value Pr(>|t|)
(Intercept)         0.6295061  0.8234732    0.764   0.4446
temperature_lag1hr -0.0500067  0.0214330   -2.333   0.0196 *
humidity_lag30min   0.0751235  0.0064451   11.656   <2e-16 ***
windspeed_lag30min  0.2322959  0.0239856    9.685   <2e-16 ***
powerdemand_lag1    0.9984154  0.0001154 8649.976   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 71.29 on 314292 degrees of freedom
Multiple R-squared:  0.9966,    Adjusted R-squared:  0.9966
F-statistic: 2.324e+07 on 4 and 314292 DF,  p-value: < 2.2e-16
```

Fig 9: Model C MLR Output in R

```
> cat("Model A (Current Temperature Only):\nRMSE =", rmse_A, ", R² =", r2_A, "\n")
Model A (Current Temperature Only):
RMSE = 79.25873 , R² = 0.9966998
> cat("Model B (Temperature Lagged 30min):\nRMSE =", rmse_B, ", R² =", r2_B, "\n")
Model B (Temperature Lagged 30min):
RMSE = 79.26638 , R² = 0.9966992
> cat("Model C (Temperaure Lagged 1hr):\nRMSE =", rmse_C, ", R² =", r2_B, "\n")
Model C (Temperaure Lagged 1hr):
RMSE = 79.26649 , R² = 0.9966992
```

Fig 10: Model Performance

	Estimate of Temp.	Pr(>|t|) of Temp.	RMSE	R^2	AIC
Model A	0.1105	0.0000	79.2587	0.9967	3574050
Model B	0.0161	0.4590	79.2663	0.9967	3574075
Model C	-0.0499	0.0199	79.2664	0.9967	3574070

Table 6: Comparison of Models A, B, and C

Very high R^2 values appeared for the three models (~0.9966) with a Residual SE that varied around 71.28–71.29 and consequently a RMSE that also did not differ too much: Model A: 79.2587, Model B: 79.2663, and Model C: 79.2664 again pointing to very similar prediction accuracy. The AIC indicated that Model A, with the smallest value (AIC=3,574,050), was the best compromise between fit and model complexity, followed by Model C and Model B.

But, while most statistically robust, Model A depend on report temperatures at the time of forecast. From another point of view, Model B, although slightly less accurate and with a non-significant temperature factor, is more practical and suited to forecasting in real applications where only historical data can be employed. Model C, which employed historical data, however, highlights an incorrect relationship and is therefore less desirable.

6. Final Model Evaluation

Model B is the most appropriate to be used when short-term demand is to be forecasted with the use of weather variables as predictors. In real-life forecasting, some of those (especially for short-term forecasts) are not available because, at prediction time, those values are not known yet or, as in time series, they are future values. Model A is not practical since the current temperature used as a predictor in this model might not be available at the time of forecasting in practice, even though the model shows the strongest statistical performance. While the temperature variable of Model B is not significant, it is still accurate (RMSE = 79.26, R^2 = 0.9967) and relies only on the lag variables, which implies that it can be used in practice and that it is theoretically consistent with the nature of causal forecasting. Model B is the best compromise between being practical for forecasting with good predictive accuracy and being developed using only information that would be available at the time of prediction.

Multiple Linear Regression Equation

The regression Equation is as follows.

$$\text{Power\_Demand} = -1.0976 + 0.0161\times(\text{Temperature\_lag30min}) + 0.0859\times(\text{Humidity\_lag30min}) + 0.2188\times(\text{Windspeed\_lag30min}) + 0.9983\times(\text{PowerDemand\_lag1})$$

The coefficients indicate the effect of a one-unit change in each independent variable on the dependent variable, assuming all other variables in the model remain constant. A 1°C increase in temperature (30 minutes earlier) is associated with a 0.0161 kW increase in power demand, holding other variables constant. A one-unit increase in humidity (30 minutes earlier) is associated with a 0.0859 kW increase in power demand, holding other variables constant. A one-unit increase in wind speed (30 minutes earlier), where wind speed is measured in meters per second (m/s), is expected to raise the power demand by 0.2188kW, given all variables remain constant. For a 1 kW rise in energy demand from the last time step, the current demand increases by 0.9983 kW.

Actual vs Predicted Power Demand

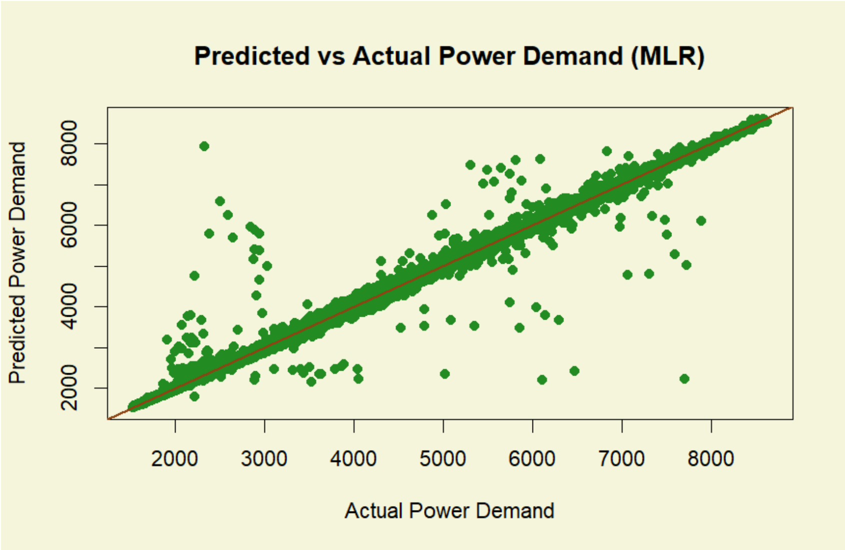

Fig 11: Actual vs Predicted Power Demand

The relationship between the actual power demand and the predicted power demand using the MLR model is shown in Figure 11. Each green dot is a single observation, and the solid brown line is the perfect 45-degree line, where we would have perfect predictions. The general trend of the data points closely compares with the 45-degree-diagonal reference line, which suggests that the model generally is good at associating the input features and the power demand. The points are evenly scattered from the line, indicating very little bias. There are some deviations, though, especially at the lower parts of the demand, meaning it might be slightly underfitted, or those parts are just more inconsistent.

Overall, the MLR model demonstrates good predictive accuracy, with most predicted values aligning closely with the actual observed power demand.

7. Conclusion

This study, conducted for short-term electricity demand prediction in Delhi, based on weather-related variables, evaluated the performance of three MLR models accounting for different forms of the temperature variable while retaining other predictors constant. Of these three models, Model A with current temperature was the most statistically significant, and best-fit model. Nevertheless, since it needs the temperature profile data at present, which is not allowed to be used in the current forecast now, it is inappropriate for practical applications.

Model B, with a temperature, lagged 30 min, is the most feasible model. Although the temperature variable was not significant in this version, Model B had high predictive capability ($R^2 = 0.9967$, RMSE = 79.26) and used only historical data, which was consistent with the situation of the short-term prediction. Model C, which included temperature lagged by 1 hour, was less preferred because of the paradoxical negative estimate for temperature.

Therefore, Model B is the optimum point compromise between precision and practicability and is best suitable for weather-based short-term real-time electricity demand prediction in Delhi.

8. Future Development

Further validation with various seasons and locations will be required to consider the robustness and applicability of the forecasting model. Electricity demand could change significantly across regions because of regional climate variations and seasonal consumption preferences; thus, the proposed model, if tested in different settings, could give us confidence in the transferability and robustness of the model. Furthermore, investigating composite forecasting models where the interpretability of statistical methods such as MLR is coupled with the prediction power of top-notch machine learning models such as Random Forests and Neural Networks may lead to more accurate forecasts. However, hybrid models might be able to capture both linear and non-linear patterns and are, hence, more responsive to complex and flexible electricity consumption behaviour.

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Appendices

```
setwd("~/R Class")


library(readr)
library(dplyr)
library(lubridate)


# Load CSV
df <- read_csv("powerdemand_5min_2021_to_2024_with weather.csv")


# rename column
df <- df %>%
  rename(
    power_demand = `Power demand`,
    temperature = temp,
    dew_point = dwpt,
    humidity = rhum,
    wind_dir = wdir,
    wind_speed = wspd,
    pressure = pres,
    moving_avg = moving_avg_3
  )



# Convert date column to Date format
df$datetime <- as.Date(df$datetime)


# Check missing values
colSums(is.na(df))


# Drop rows with any NA
df <- na.omit(df)
```

```
# 5-minute data: 6 lags = 30 min, 12 lags = 1 hour
# Create lagged variables using dplyr::lag()
df <- df %>%
  mutate(
    temperature_lag30min = lag(temperature, 6),
    temperature_lag1hr = lag(temperature, 12),
    humidity_lag30min = lag(humidity, 6),
    humidity_lag1hr = lag(humidity, 12),
    windspeed_lag30min = lag(wind_speed, 6),
    windspeed_lag1hr = lag(wind_speed, 12)
  )

# Check missing values
colSums(is.na(df))

# Drop rows with new NA values (from lagging)
df <- df %>%
  filter(
    !is.na(temperature_lag30min),
    !is.na(temperature_lag1hr),
    !is.na(humidity_lag30min),
    !is.na(humidity_lag1hr),
    !is.na(windspeed_lag30min),
    !is.na(windspeed_lag1hr)
  )

### Descriptive Statistics
summary(df)

#Checking skewness of the data
install.packages("e1071")  # Run this only once
library(e1071)
```

```
}

# Calculate skewness for all numeric variables

numeric_vars <- sapply(df, is.numeric)
skew_values <- sapply(df[, numeric_vars], skewness, na.rm = TRUE)

# View skewness

print(skew_values)


#Box plot for each numeric variable
library(ggplot2)
library(tidyr)

# List of numeric columns (modify as needed)
numeric_vars <- c("power_demand", "temperature", "humidity", "wind_speed",
         "temperature_lag30min", "temperature_lag1hr",
         "humidity_lag30min", "humidity_lag1hr",
         "windspeed_lag30min", "windspeed_lag1hr",
         "pressure", "moving_avg")

# Plot individually
for (var in numeric_vars) {
 print(
  ggplot(df, aes_string(y = var)) +
   geom_boxplot(fill = "skyblue", outlier.color = "red") +
   ggtitle(paste("Boxplot of", var)) +
   theme_minimal()
 )
}

# Linearity & Homoscedasticity Assumption
```

```
# Fit basic MLR model
model <- lm(power_demand ~ temperature + humidity + wind_speed +
            temperature_lag30min + temperature_lag1hr +
            humidity_lag30min + humidity_lag1hr +
            windspeed_lag30min + windspeed_lag1hr +
            pressure + moving_avg, data = df)

# Residual plot
plot(model$fitted.values, model$residuals,
    xlab = "Fitted Values", ylab = "Residuals",
    main = "Residual Plot")
abline(h = 0, col = "red")

data <- data.frame(
  Predicted = model$fitted.values,
  Residuals = model$residuals
)

ggplot(data, aes(x = Predicted, y = Residuals)) +
  geom_point(color = "#228B22", alpha = 0.7, size = 1.2) +  # Forest green
  geom_hline(yintercept = 0, color = "#8B4513", size = 1) +  # Saddle brown
  labs(title = "Residual Plot", x = "Predicted power_demand", y = "Residuals") +
  theme_minimal(base_size = 14) +
  theme(
    plot.background = element_rect(fill = "#f5f5dc", color = NA),  # Beige
    panel.background = element_rect(fill = "#f5f5dc", color = NA),
    panel.grid.major = element_line(color = "#d2b48c"),  # Tan grid
    panel.grid.minor = element_blank(),
    axis.title = element_text(color = "black"),
    plot.title = element_text(hjust = 0.5)
  )
```

```
#Autocorrelation / Independence of Residuals (Durbin-Watson Test)
library(lmtest)
dwtest(model)  # p > 0.05 means no autocorrelation (good)

#fix autocorrelation
# Include Lagged Dependent Variables (Autoregressive Terms)
df <- df %>%
  mutate(
    powerdemand_lag1 = lag(power_demand, 1),
    powerdemand_lag2 = lag(power_demand, 2)
  ) %>%
filter(!is.na(powerdemand_lag1), !is.na(powerdemand_lag2))

model_ar <- lm(power_demand ~ temperature + humidity + wind_speed +
                 temperature_lag30min + temperature_lag1hr +
                 humidity_lag30min + humidity_lag1hr +
                 windspeed_lag30min + windspeed_lag1hr +
                 powerdemand_lag1 + powerdemand_lag2,
               data = df)

library(lmtest)
dwtest(model_ar)

##Autocorrelation has been resolved

#Residual Plots for Linearity & Homoscedasticity
# Residual vs Fitted plot
plot(model_ar, which = 1)
abline(h = 0, col = "red")

# Normal Q-Q plot
plot(model_ar, which = 2)
```

```
#Histogram of Residuals
hist(residuals(model_ar), breaks = 50, col = "lightblue", main = "Histogram of Residuals")

library(ggplot2)
ggplot(data.frame(resid = residuals(model_ar)), aes(x = resid)) +
  geom_histogram(bins = 50, fill = "skyblue", color = "black") +
  labs(title = "Histogram of Residuals")

#Check for Multicollinearity (VIF)
library(car)
vif(model_ar)

#Fixed the multicolinearity
#Calculate correlations between lagged variables and power demand
# Load required libraries
library(corrplot)
install.packages("ggcorrplot")
library(ggcorrplot)

# Select only numeric columns relevant for correlation check
cor_data <- df[, c("power_demand",
               "temperature", "temperature_lag30min", "temperature_lag1hr",
               "humidity", "humidity_lag30min", "humidity_lag1hr",
               "wind_speed", "windspeed_lag30min", "windspeed_lag1hr")]

# Compute correlation matrix
corr_matrix <- cor(df[, c("power_demand", "temperature", "temperature_lag30min",
                    "temperature_lag1hr", "humidity", "humidity_lag30min",
                    "humidity_lag1hr", "wind_speed", "windspeed_lag30min",
                    "windspeed_lag1hr")], use = "complete.obs")

# Visualize the correlation heatmap
corrplot(cor_matrix,
```

```
        method = "color",          # Color tiles
        type = "lower",            # Show only lower triangle
        tl.col = "black",          # Label color
        tl.cex = 0.7,
        tl.srt = 90,               # Rotate labels
        addCoef.col = "black",     # Add correlation coefficients
        number.cex = 0.7,          # Coefficient font size
        number.digits = 2,
        col = colorRampPalette(c("blue", "white", "red"))(200),  # Color gradient
        title = "Correlation Matrix", # Add title
        mar = c(0,0,1,0))

#Correlation between powerdemand lag
cor(df[, c("power_demand", "powerdemand_lag1", "powerdemand_lag2")], use = "complete.obs")

#Create MLR Model with weather lagged variables
# Split data (e.g., 80-20 train-test)
set.seed(123)
n <- nrow(df)
train_index <- 1:floor(0.8 * n)
train_data <- df[train_index, ]
test_data <- df[-train_index, ]

#Only Weather Lagged Variables
model_lagged <- lm(power_demand ~ temperature_lag30min + humidity_lag30min +
windspeed_lag30min +
           powerdemand_lag1, data = train_data)
#summary
summary(model_lagged)

#Predict on Test Set
# Predictions
pred_lagged <- predict(model_lagged, newdata = test_data)
```

```
# Actual values
actual <- test_data$power_demand

#Calculate RMSE and R²
# Load Metrics Library
install.packages("Metrics")
library(Metrics)

#Model Perdormance
rmse_lagged <- rmse(actual, pred_lagged)
r2_lagged <- 1 - sum((actual - pred_lagged)^2) / sum((actual - mean(actual))^2)

# Display
cat("Model with only Lagged Weather):\nRMSE =", rmse_lagged, ", R² =", r2_lagged, "\n")

#Compare model with different temperature lag
#Model A: current temperature
model_A <- lm(power_demand ~ temperature + humidity_lag30min + windspeed_lag30min +
        powerdemand_lag1, data = train_data)

#Model B: temperature_lag30min
model_B <- lm(power_demand ~ temperature_lag30min + humidity_lag30min + windspeed_lag30min +
        powerdemand_lag1, data = train_data)

#Model C: temperature_lag1hr
model_C <- lm(power_demand ~ temperature_lag1hr + humidity_lag30min + windspeed_lag30min +
        powerdemand_lag1, data = train_data)

#summary of the model
summary(model_A)
summary(model_B)
summary(model_C)
```

```
#Predict on Test Set
# Predictions
pred_A <- predict(model_A, newdata = test_data)
pred_B <- predict(model_B, newdata = test_data)
pred_C <- predict(model_C, newdata = test_data)

# Actual values
actual <- test_data$power_demand

#Calculate RMSE and R²
# Load Metrics Library
install.packages("Metrics")
library(Metrics)

# Model A Performance
rmse_A <- rmse(actual, pred_A)
r2_A <- 1 - sum((actual - pred_A)^2) / sum((actual - mean(actual))^2)

# Model B Performance
rmse_B <- rmse(actual, pred_B)
r2_B <- 1 - sum((actual - pred_B)^2) / sum((actual - mean(actual))^2)

#Model C Perdormance
rmse_C <- rmse(actual, pred_C)
r2_C <- 1 - sum((actual - pred_C)^2) / sum((actual - mean(actual))^2)

# Display
cat("Model A (Current Temperature Only):\nRMSE =", rmse_A, ", R² =", r2_A, "\n")
cat("Model B (Temperature Lagged 30min):\nRMSE =", rmse_B, ", R² =", r2_B, "\n")
cat("Model C (Temperaure Lagged 1hr):\nRMSE =", rmse_C, ", R² =", r2_B, "\n")

#AIC comparison
```

```r
AIC(model_A, model_B, model_C)


# Plot Actual vs Predicted values
par(bg = "#f5f5dc")
# Create the plot
plot(test_data$power_demand, pred_B,
    xlab = "Actual Power Demand",
    ylab = "Predicted Power Demand",
    main = "Predicted vs Actual Power Demand (MLR)",
    col = "#228B22", pch = 16,
    panel.first = rect(par("usr")[1], par("usr")[3],
                  par("usr")[2], par("usr")[4],
                  col = "#f5f5dc", border = NA))  # Panel background


# Add 45-degree reference line
abline(a = 0, b = 1, col = "#8B4513", lwd = 2)
```